

KANNAN APPIYA SANTHARAM

Senior Lead Software Engineer (Frontend/Full Stack) | React · TypeScript · Node.js · AI-Native

Chennai, India

+91 97902 47499 | as.kannan4@gmail.com | kannansantharam.com/ind
linkedin.com/in/askannan | github.com/kannan-santharam

Nationality: Indian | **Notice Period:** 60 Days

PROFESSIONAL SUMMARY

Full Stack Senior Lead Software Engineer with 10.5+ years in enterprise SaaS, working across React, TypeScript, and Node.js. Currently leading an engineering team on a platform serving 4,000+ MSP and IT enterprise customers, driving technical direction across AI-native engineering with LLM orchestration, MCP servers, and production agent skills. Seeking a Senior Lead / Staff Engineer or Engineering Manager role.

KEY ACHIEVEMENTS

- **AI test-authoring platform** that generates, runs, and self-heals a 300+ test-case Playwright suite, reducing manual QA effort.
- **Headless Base UI design system** with an AI agent that generates production components from a Figma link, taking page development from 2 days of AI-assisted work to a few hours.
- **96% faster dev builds**, 2 minutes to 5 seconds across 12 monorepo packages.
- **70% faster initial page load**, 5 seconds down to 1.5 seconds.
- **Mentored 10+ engineers**, junior to mid-level, across SuperOps and Freshworks.

SKILLS

Full Stack Engineering: React, Next.js, TypeScript, JavaScript (ES6+), HTML5 & CSS3, Responsive Web Design, Tailwind CSS, Internationalisation (i18n, react-intl), Node.js, MySQL, Python, REST API Design & Integration, Microservices, GraphQL & Apollo, Zustand, Unit Testing (Jest, Vitest), Playwright, Jenkins CI/CD, Docker, AWS

Leadership & Delivery: Engineering Team Leadership, Solution Architecture & System Design, Developer Mentorship (10+ engineers), Architecture & Code Reviews, Agile & Scrum Delivery, Cross-Team Collaboration, Stakeholder Management & Customer Communication

AI-Native Engineering: Production Agent Skills, Model Context Protocol (MCP) Servers, LLM Orchestration, AI Agent Guardrails & Safety, RAG & Embedding, Amazon Bedrock AgentCore

Platform & Performance: Monorepos, Module Federation, Rspack & Webpack 5, Micro-Frontends, Design Systems & UI SDKs, Performance Optimisation & Core Web Vitals

Tools: Claude Code, Cursor, GitHub Copilot, Figma, Knip Static Analysis

WORK EXPERIENCE

SUPEROPS

Senior Lead Software Engineer

Chennai, India

Jul 2022 – Present

Progression: Senior Software Engineer to Lead Software Engineer to Senior Lead Software Engineer

- Hands-on delivery of enterprise features in React and TypeScript, including role-based access control (RBAC) and near-real-time operational dashboards tuned with custom cached Apollo hooks to balance data freshness against network load, in partnership with product and UI/UX teams.
- Lead and manage a frontend team of 4 to 6 engineers on a platform serving 4,000+ MSP and IT enterprise customers, owning sprint planning, architecture reviews, hiring and interviews, coaching and performance feedback, career development, and roadmap prioritisation with product managers.
- Reduced dev build time by 96%, from 2 minutes to 5 seconds, through a solo Webpack 5 to Rspack migration delivered in 3 weeks across 12 monorepo packages, saving hundreds of engineering hours per month and accelerating time-to-market for every downstream feature.
- Reduced manual regression effort and release risk by architecting an AI test-authoring platform in React and Node.js that orchestrates LLM agents to generate, execute, and self-heal end-to-end test suites, streaming real-time results over HTTP Web Streams, producing a 300+ test-case Playwright regression suite run in Jenkins CI/CD.
- Improved developer productivity by integrating custom Claude Code skills and MCP servers into the

production development workflow, with multi-tier write-scope guardrails that let engineers run autonomous agents without exposing core platform interfaces to risk.

- Built a Python MCP server exposing the frontend codebase as a queryable graph, now used by AI tooling across multiple engineering teams.
- Built a headless Base UI design system on customised design tokens and a Claude Code skill that generates production components from a Figma node link via MCP, reducing page development from 2 days of AI-assisted engineering to a few hours, alongside a reusable UI SDK adopted across product teams.
- Cut initial page load from 5 seconds to 1.5 seconds through route-level code splitting, React lazy loading (Suspense), and per-package bundle chunking, and eliminated 30,000+ lines of dead code and unused dependencies to reduce technical debt and the build and security surface.

FRESHWORKS

Senior Software Engineer (Customer-Facing Engineering)

Chennai, India
Jun 2018 – Jul 2022

Progression: Onboarding Engineer to Senior Software Engineer

- Acted as the technical point of contact for enterprise customers, partnering with Support, Customer Success, and Sales to turn complex account requirements into solutions that unblocked onboarding and renewals, recognised with the Customer Champion Award.
- Reduced support handling effort by building a Customer 360 dashboard consolidating support, sales, and account data into a single operational view.
- Integrated tier-1 SaaS systems including Jira, Salesforce, Zendesk, and ServiceNow through REST API services, expanding platform fit in enterprise deals.
- Developed features end-to-end across Freshworks products, building the React UI with the Node.js REST API services and MySQL data layer behind it, plus reusable add-ons that met customer-specific requirements without one-off custom development.
- Mentored 6 to 7 junior and onboarding engineers on React, REST API development, and UI architecture through structured code and design reviews, accelerating team onboarding.

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Software Developer (acquired by Freshworks)

Chennai & Bengaluru, India
Mar 2016 – Jun 2018

Progression: Web Application Developer & Designer to Software Developer

- Built an admin dashboard that dynamically priced Stripe plans across feature selection, storage tiers, and bandwidth limits, increasing self-serve revenue.
- Built video player customisation from scratch with configurable controls, annotations, and call-to-action overlays, increasing viewer conversion.
- Built responsive, high-converting company and product landing pages.
- Integrated Freshservice with DocuSign over REST API services to extend enterprise workflow coverage, joining Freshworks with the team on acquisition.

EDUCATION

Bachelor of Engineering (B.E.), Computer Science

2011 – 2015

K.L.N. College of Information Technology (Affiliated with Anna University), Tamil Nadu, India

LANGUAGES

English: Fluent (Professional Working Proficiency) | **Tamil:** Native | **Hindi:** Intermediate